# SIMATS ENGINEERING
## ASSESSMENT TOOL 1: Concept Mapping

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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**COURSE OUTCOME COVERED**
**CO1:** *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 4

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**Code of Conduct**
I Bala s (192524427) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: bala

**Assessment Tasks**
**Concept Mapping Tasks**
Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

**Expected concept nodes:**
- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

**Concept Mapping Rubric**

| Criteria | Weightage | Level 4 - Exemplary | Level 3 - Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Concept Identification | 25% | All major concepts are accurately identified | Most concepts are identified correctly | Limited concepts are identified | Essential concepts are missing |
| Relationship Mapping | 25% | Links between concepts are accurate and meaningful | Most links are correct | Some links are weak or unclear | Relationships are mostly incorrect |
| Hierarchy and Organization | 20% | Excellent structure with clear hierarchy | Mostly organized with minor issues | Basic structure with weak hierarchy | No logical organization |
| Technical Relevance | 20% | Map strongly reflects cloud course content | Mostly aligned to topic | Partially aligned to topic | Weak alignment to topic |
| Visual Clarity | 10% | Neat, readable, and easy to interpret | Generally readable | Somewhat cluttered | Poorly presented |

5. Questions

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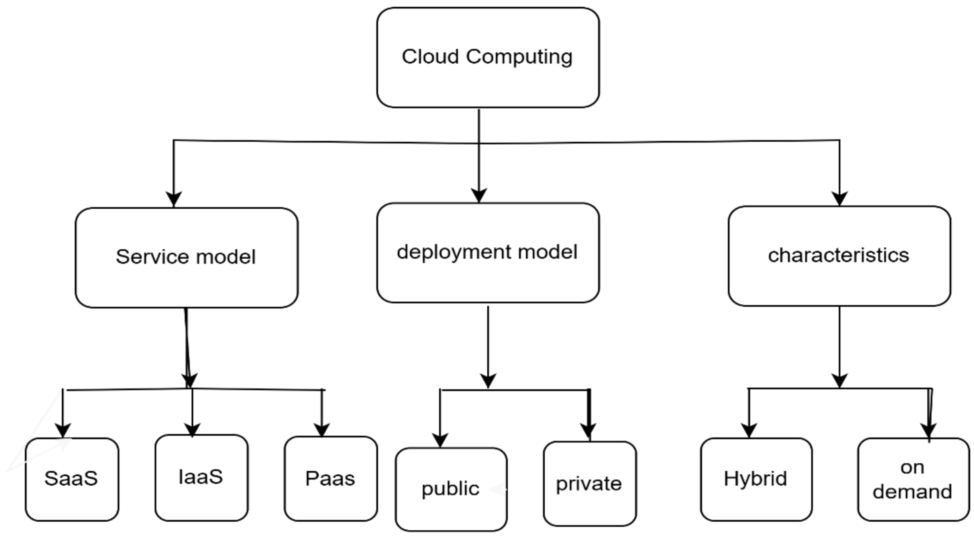
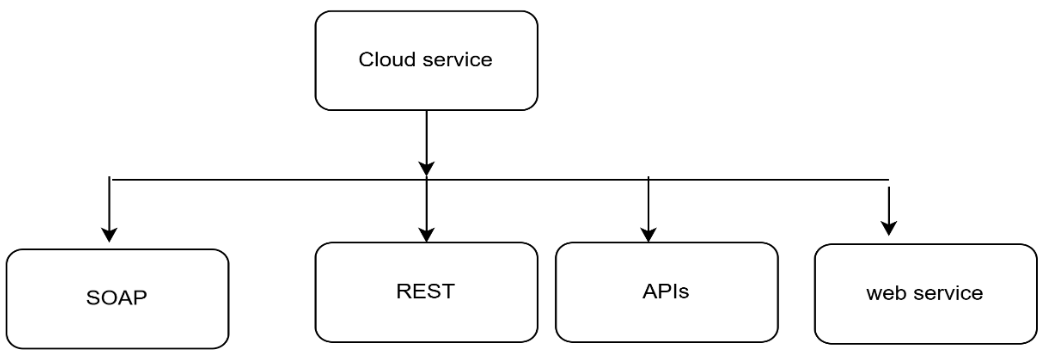
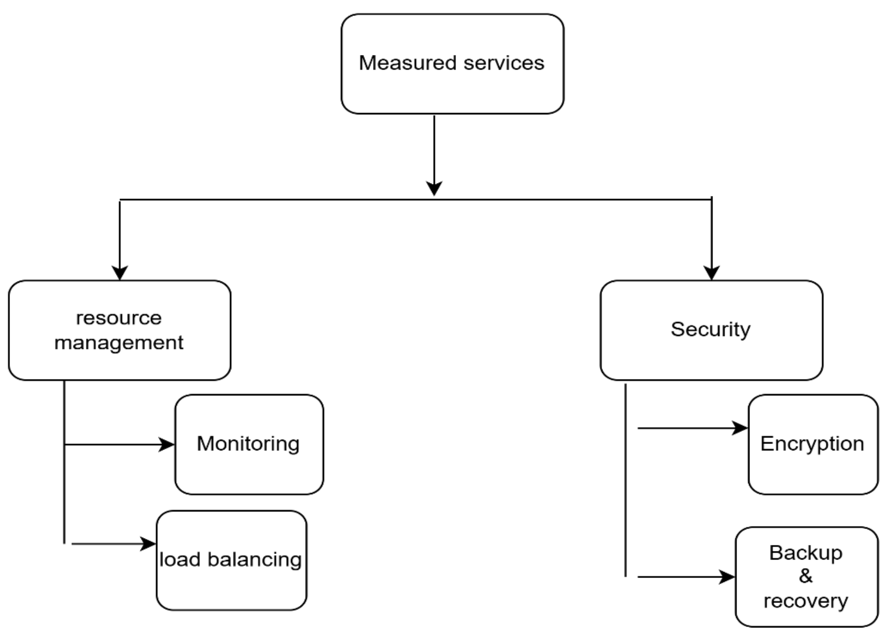
| i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why? |
| ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent? |
| iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system? |
| iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience? |
| v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it? |

Concept maps

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- **The concept map effectively represents cloud computing principles by organizing the main concepts into a clear hierarchical structure.**
- **The central idea, Cloud Computing, branches into key areas such as Service Models (IaaS, PaaS, SaaS), Deployment Models (Public, Private, Hybrid, Community)**
- **Core Characteristics (on-demand self-service, resource pooling, rapid elasticity, measured service, and broad network access), Benefits, Challenges, and Security**
- **. This structure makes it easy to understand the relationships between concepts and provides a logical flow of information**
- **The most important design decision was placing Cloud Computing at the center and connecting all major concepts to it.**
- **This creates a clear overview and helps users understand how each component contributes to the overall cloud computing ecosystem.**
- **A second key decision was grouping related concepts into categories rather than listing them individually.**
- **For example, keeping all service models together allows for quick comparison and highlights their differences and similarities.**
- **Another effective decision was using connecting lines and descriptive labels to show relationships such as "includes," "provides," and "depends on," making the connection.**

**2.**

- **Scalability, elasticity, and resource management are fundamental principles that shape the architecture of cloud computing. Scalability enables the system to increase or decrease computing resources, such as servers, storage, and processing power, to meet growing or changing user demands. This ensures that applications maintain good performance as workloads increase.**
- **Resource management is responsible for allocating, monitoring, and optimizing shared cloud resources. It ensures that computing resources are distributed efficiently among users, prevents resource conflicts, and supports reliable performance through techniques such as virtualization, load balancing, and monitoring.**
- **If these capabilities were absent, several challenges would arise:**
- **Applications could become slow or unavailable during periods of high demand because resources could not scale.**
- **Organizations would either over-provision resources (leading to higher costs) or under-provision them (causing poor performance).**
- **Manual resource allocation would increase administrative effort and reduce operational efficiency.**
- **Shared resources could be used inefficiently, resulting in resource contention, downtime, and reduced reliability.**
- **Overall user experience and service availability would decline, making cloud computing less flexible and less cost-effective.**

- **In summary, scalability, elasticity, and resource management are essential to cloud architecture because they ensure high performance, efficient resource utilization, cost optimization, and reliable service delivery.**

**3.**

- **The APIs, web services, and cloud services included in the concept map were selected because they are essential components that enable communication, integration, and efficient delivery of cloud-based applications.**

- **APIs (Application Programming Interfaces) were included because they allow different applications and services to communicate and exchange data securely. APIs enable developers to integrate cloud resources such as storage, databases, authentication, and payment services into applications without building these features from scratch. This improves development speed, flexibility, and interoperability.**

- **Web services were selected because they provide standardized methods for applications to communicate over the internet using protocols such as REST and SOAP. They enable data sharing between different platforms and support distributed systems, making cloud applications more reliable and compatible across various devices and operating systems.**

- **Cloud services such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) were included because they represent the core service models of cloud computing. IaaS provides virtualized computing resources, PaaS offers a platform for developing and deploying applications, and SaaS delivers ready-to-use software over the internet. Together, these services reduce infrastructure costs, improve scalability, and simplify application deployment and maintenance.**

- **Overall, these components enhance the system by enabling seamless communication, efficient resource utilization, rapid application development, and scalable service delivery. Their integration ensures that cloud applications remain flexible, reliable, secure, and cost-effective while meeting the needs of both users and organizations.**

**4.**

- **The relationships between the components in the concept map demonstrate how different cloud computing elements work together to provide a reliable, scalable, and efficient system. The service models (IaaS, PaaS, and SaaS) depend on the underlying cloud infrastructure, with each layer building on the services provided by the one below it. This layered architecture simplifies application development and deployment while reducing the complexity for end users.**

- **The interaction between scalability, elasticity, and resource management ensures that resources are allocated efficiently according to demand. Resource management continuously monitors workloads, while elasticity automatically adjusts resources to maintain performance. This relationship improves reliability by reducing downtime, enhances performance by preventing resource bottlenecks, and optimizes costs by avoiding unnecessary resource allocation.**

- **APIs and web services connect cloud applications with external systems, enabling seamless data exchange and interoperability. These interactions improve the user experience by providing faster access to services, supporting integration across multiple platforms, and**

**ensuring consistent application functionality. In addition, the relationship between security services, storage, and network infrastructure protects user data while maintaining high availability and secure access to cloud resources.**

- **Overall, the strong connections between these components create a cloud architecture that is flexible, resilient, and capable of delivering high-performance services with minimal disruption.**

**5.**

- **Developing the concept map improved my understanding of cloud computing architecture by showing how individual concepts are interconnected rather than existing independently. Instead of viewing service models, deployment models, security, scalability, and resource management as separate topics, I gained a clearer understanding of how they work together to deliver cloud services efficiently.**
- **The process also highlighted the importance of organizing information logically. Placing Cloud Computing at the center and grouping related concepts into categories made it easier to visualize the relationships and understand how each component contributes to the overall architecture. Analyzing these connections reinforced the role of APIs, web services, and cloud services in supporting communication, scalability, and system integration.**
- **If I were redesigning the concept map, I would improve it by adding more detailed relationship labels to explain how components interact, including real-world examples such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). I would also use clearer visual distinctions, such as color coding and directional arrows, to better illustrate data flow, dependencies, and interactions between components. These improvements would make the concept map easier to interpret and provide a more comprehensive representation of cloud computing architecture.**